Lipid Exudation Following Plaque Radiotherapy for Posterior Uveal Melanoma

Michael D. Mills, MD, and
J. William Harbour, MD

PURPOSE: **To investigate factors associated with lipid exudation after plaque radiotherapy for uveal melanoma.**
DESIGN: **Retrospective interventional case series.**
METHODS: **Records of 139 consecutive uveal melanoma patients treated with I-125 plaque radiotherapy were reviewed. Lipid exudation was the major outcome measured.**
RESULTS: **Lipid exudation was detected in 19 patients (13.7%). Mean time from plaque therapy to lipid exudation was 10 months (range, 3–23 months). Lipid exudation was associated with younger patient age (P = .0002), increased low-density lipoprotein (P = .004), decreased high-density lipoprotein (P = .026), increased tumor thickness (P = .024), and exudative retinal detachment (P = .016). Patients with lipid exudation had worse visual outcome ($P < .0001$) and increased neovascular glaucoma (P = .01).**
CONCLUSIONS: **Lipid exudation after plaque radiotherapy for uveal melanoma is associated with poor ocular outcome. Factors associated with this lipid exudation are distinct from those associated with radiation retinopathy, and they may suggest interventions to combat this complication. (Am J Ophthalmol 2006;141:594–595. © 2006 by Elsevier Inc. All rights reserved.)**

PLAQUE RADIOTHERAPY IS A COMMON TREATMENT FOR uveal melanoma.[1] Ocular outcomes after plaque radiotherapy are dependent on many factors, including lipid exudation from the irradiated tumor.[2] Unfortunately, little is known about the risk factors for this widely recognized complication. We reviewed 139 consecutive uveal melanoma patients after plaque radiotherapy to identify the factors associated with lipid exudation.

Institutional Review Board approval was obtained. Consecutive patients with posterior uveal melanoma treated with I-125 plaque radiotherapy that used Collaborative Ocular Melanoma Study (COMS) guidelines between August 1996 and September 2003 were included. Patients were excluded from statistical analysis if there was no follow-up (two patients) or if photographs or drawings were not available to assess lipid exudation (two patients).

Recorded data included dates of plaque therapy and final follow-up, patient age, gender, ethnicity, presence or absence of diabetes mellitus and hypertension, use of lipid-lowering agents, involved eye, Snellen visual acuity, tumor location, proximity to optic disk and fovea, pigmentation, largest basal tumor dimension, A-scan thickness, collar button, subretinal or vitreous hemorrhage, exudative retinal detachment, neovascular glaucoma, adjuvant transpupillary thermotherapy, secondary enucleation, local recurrence, and systemic metastasis. Lipid exudation was classified as absent, mild (thin and within 5 mm of tumor), moderate (thick and within 5 mm of tumor), or severe (marked and extending more than 5 mm beyond tumor). Recorded radiotherapy parameters included plaque size, treatment time, and total radiation dose to tumor apex and base. Where available, serum triglycerides, total cholesterol, low-density lipoprotein, and high-density lipoprotein were recorded.

Comparison of continuous variables was performed by the Student t test, categorical variables by χ^2 analysis, and time-dependent variables by Kaplan-Meier life table analysis (MedCalc software, version 7.2.0.2).

The study included 139 patients with posterior uveal melanoma with tumor size graded by COMS criteria as small in eight patients, medium in 123 patients, and large in eight patients. All patients were white, and all tumors were unilateral. Of these patients, 19 (13.7%) developed lipid exudation, which was graded as mild in seven patients, moderate in six patients, and severe in six patients (Figure 1). Mean time from plaque therapy to onset of lipid exudation was 10 months (median 9 months; range 3–23 months) (Figure 2). Final follow-up occurred at a mean of 37 months (median 34 months) after plaque therapy. Factors significantly associated with lipid exudation are summarized in the Table.

Almost 16% of uveal melanoma patients treated with I-125 plaque radiotherapy developed lipid exudation, which was associated with worse visual outcome and neovascular glaucoma. To our knowledge, this is the first study to show an association between lipid exudation and serum lipid abnormalities. Interestingly, the factors associated with lipid exudation are different than the known risk factors for radiation retinopathy (for example, diabetes mellitus and high radiation dose rate),[3] suggesting that lipid exudation may not be a manifestation of radiation retinopathy but may have a distinct pathogenesis. We speculate that lipid exudation may be due to radiation-induced incompetence of tumor blood vessels aggravated by serum lipid abnormalities, analogous to lipemia retinalis.[4] The association of lipid exudation with exudative retinal detachment and increased tumor thickness may reflect more extensive tumor vasculature in these cases. Importantly, there were no significant differences between eyes with and without lipid exudation with respect to radiation dose, dose rate, or adjuvant transpupillary thermotherapy, suggesting that our findings are not due to differences in treatment.

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From the Ocular Oncology Service, Department of Ophthalmology and Visual Sciences, Washington University School of Medicine and Barnes Retina Institute, St Louis, Missouri.

Inquiries to William Harbour, MD, Department of Ophthalmology and Visual Sciences, Washington University School of Medicine, Campus Box 8096, 660 South Euclid Avenue, St Louis, MO 63110; e-mail: harbour@wustl.edu

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